

MAKEit - ISTE GCEK STUDENT CHAPTER HARDWARE HACKATHON

Rules And Regulations

Assembly & Venue Requirements

- Circuit assembly and presentations must be completed at the venue.
- Presentations can be in any format - not necessarily PowerPoint. You may use Obsidian, markdown, or any other documentation format you prefer.

Component Usage

- Only components listed in your submitted BOM may be used in your circuit or prototype. Using any unlisted component will result in **disqualification**.
- If you wish to change any component:
 - Present the new component to the organizers
 - Specify which item you want to replace
 - Update your BOM accordingly
 - Ensure it still fits within the 5-sensor limit

Problem Statement Selection

- You must choose one of the 5 problem statements provided.
- The selected problem statement must be clearly mentioned during your presentation.

Penalties

- **Budget violation:** If your total BOM exceeds the ₹1500 limit, **5 points will be deducted for every ₹25 over the limit.**
- **Sensor limit violation:** If you use more than 5 sensors, **10 points will be deducted per additional sensor.**

Judging Process

- You must answer any questions asked by the judges during evaluation.
- Teams will be evaluated based on the following criteria:

Criteria	Maximum Marks
Innovation & Creativity	25
Functionality	25
Practical Impact	20
Budget & Constraint Adherence	15
Presentation & Demo	15
Total	100

Competition Timeline

- **Total Duration:** 6 hours (10:00 AM - 4:00 PM)
- **Includes:** 1-hour lunch break
- **Final 30 minutes:** Reserved for judge evaluation and results

Final Notes

- Penalties will be applied after initial scoring
- The team with the highest final score will be declared the winner
- All decisions made by the organizers are final

DETAILED PROBLEM STATEMENTS

1. ResQTech – Emergency Detection and Alert Systems



Problem Statement

Design an intelligent emergency detection system that can automatically identify potential life-threatening situations and provide immediate alerts to save lives and prevent property damage.

Real-World Context

Every year, thousands of lives are lost due to delayed emergency response. Your system should detect emergencies like fires, gas leaks, medical emergencies, or accidents and provide instant alerts to users, family members, or emergency services.

Expected Deliverables

- ☐ Automatic hazard detection with at least 2 different sensors
- ☐ Multi-level alerting (local alarm + remote notification)
- ☐ Battery backup consideration for power failures
- ☐ User-friendly interface for configuration and monitoring

Hardware Suggestions

MQ2/MQ135 (gas) • DHT11 (temperature) • vibration sensor • buzzer • LED indicators • ESP32/Arduino

Success Criteria

🕒 System should detect emergencies within 10 seconds and trigger alerts within 15 seconds

2. GuardianNode – Smart Home Security Systems 🏠

Problem Statement

Develop an affordable yet effective smart security system that protects homes from intrusions while providing real-time monitoring and control capabilities to homeowners.

Real-World Context

Traditional security systems are expensive and require professional installation. Your system should provide robust security features at a fraction of the cost while being easy to install and manage.

Expected Deliverables

- ☐ Multi-point monitoring system covering at least 3 entry points
- ☐ User authentication mechanism (RFID, keypad, or app-based)
- ☐ Real-time notifications to homeowner's device
- ☐ Tamper detection and backup communication methods

Hardware Suggestions

PIR motion sensor • magnetic door switches • RFID/keypad module • servo motor • buzzer • ESP32/Arduino

Success Criteria

 **System should detect intrusions within 3 seconds and send alerts within 10 seconds**

3. HealthBridge – Affordable Health and Wellness Monitoring

Problem Statement

Create a comprehensive health monitoring device that tracks vital signs and wellness metrics, making healthcare monitoring accessible and affordable for everyone, especially in rural or underserved communities.

Real-World Context

Healthcare is expensive and often inaccessible. Your device should provide continuous health monitoring, early warning systems for health issues, and promote preventive healthcare practices.

Expected Deliverables

- ☐ Continuous monitoring of at least 2 health parameters
- ☐ Abnormality detection with customizable thresholds
- ☐ Data visualization and historical tracking
- ☐ Emergency alert system for critical health events

Hardware Suggestions

Pulse sensor • accelerometer • temperature sensor • RTC module • LCD/OLED display • buzzer

Success Criteria

 **System should provide accurate readings within $\pm 5\%$ of medical-grade devices**

4. AgroAssist – Smart Farming and Resource Optimization

Problem Statement

Design an intelligent agricultural monitoring and automation system that helps farmers optimize crop yield while conserving water and energy resources, especially targeting small-

scale and resource-constrained farmers.

Real-World Context

Modern farming requires precision and efficiency to feed growing populations while conserving resources. Your system should automate farming decisions, reduce manual labor, and maximize crop productivity.

Expected Deliverables

- ☐ Multi-parameter monitoring (soil moisture, temperature, humidity, light)
- ☐ Automated irrigation system with smart scheduling
- ☐ Threshold-based alerts for critical conditions
- ☐ Historical data logging for farming pattern analysis

Hardware Suggestions

Soil moisture sensor • DHT11 • LDR • water pump with relay • RTC • ESP32/Arduino

Success Criteria

💧 System should reduce water consumption by 30% while maintaining or improving crop health indicators

5. EcoTrack – Environmental Sensing and Monitoring



Problem Statement

Build a comprehensive environmental monitoring system that tracks pollution levels, resource consumption, and environmental health indicators to promote sustainable living and environmental awareness in communities.

Real-World Context

Environmental degradation affects everyone, but many people lack awareness of their environmental impact. Your system should provide real-time environmental data and encourage sustainable practices.

Expected Deliverables

- ☐ Real-time environmental sensing of at least 3 parameters
- ☐ Data visualization with easy-to-understand metrics

- ☐ Threshold-based warnings for dangerous levels
- ☐ Sustainability recommendations based on collected data

Hardware Suggestions

MQ135 (air quality) • sound sensor • current sensor • ultrasonic sensor • DHT11 • ESP32/Arduino

Success Criteria

 System should provide environmental readings accurate enough to guide personal or community decisions



Quick Reference

Problem Track	Primary Focus	Key Sensors	Target Impact
ResQTech	Emergency Detection	Gas, Temperature, Vibration	Life Safety
GuardianNode	Home Security	PIR, Door Switch, RFID	Property Protection
HealthBridge	Health Monitoring	Pulse, Motion, Temperature	Healthcare Access
AgroAssist	Smart Farming	Soil, Climate, Light	Resource Efficiency
EcoTrack	Environmental	Air Quality, Sound, Energy	Sustainability